

Erdős Problem #33

EP-33

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $A \subset \mathbb{N}$ be such that every large integer can be written as $n^2 + a$ for some $a \in A$ and $n \geq 0$. What is the smallest possible value of $\limsup \frac{|A \cap \{1, \dots, N\}|}{N^{1/2}}$? Is $\liminf \frac{|A \cap \{1, \dots, N\}|}{N^{1/2}} > 1$?

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.
Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>